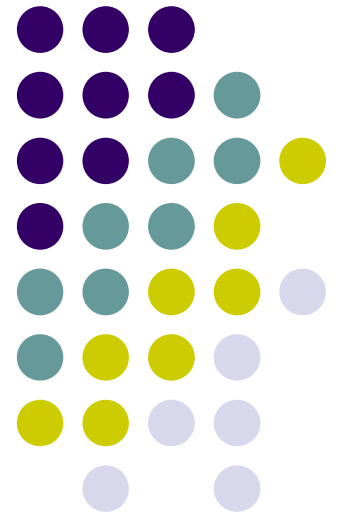
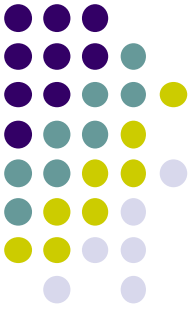


WIA1005 Network Technology Foundation

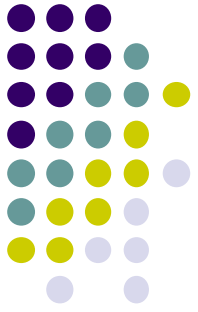
Chapter 4 Network and Transport Layer





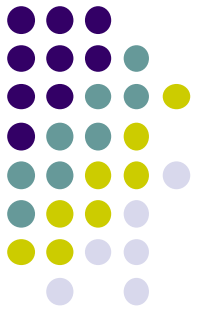
Contents

- Network Layer
- Routing
- ICMPv4
- Transport Layer
- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)



Network Layer

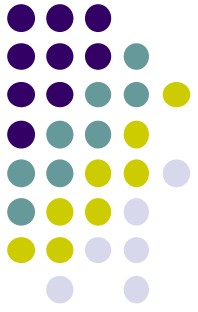
- The network layer allow end devices to exchange data across the network. It consists of four basic processes:
 - **Addressing end devices** - End devices must be configured with a unique **IP address** for identification on the network.
 - **Encapsulation** - The network layer adds IP header to the PDU receive from the transport layer.
 - **Routing** - The network layer provides services to **direct packets** to a destination host on another network.
 - **De-encapsulation** - When the packet arrives at the destination host, the IP header is removed from the packet. and passed up to the transport layer.



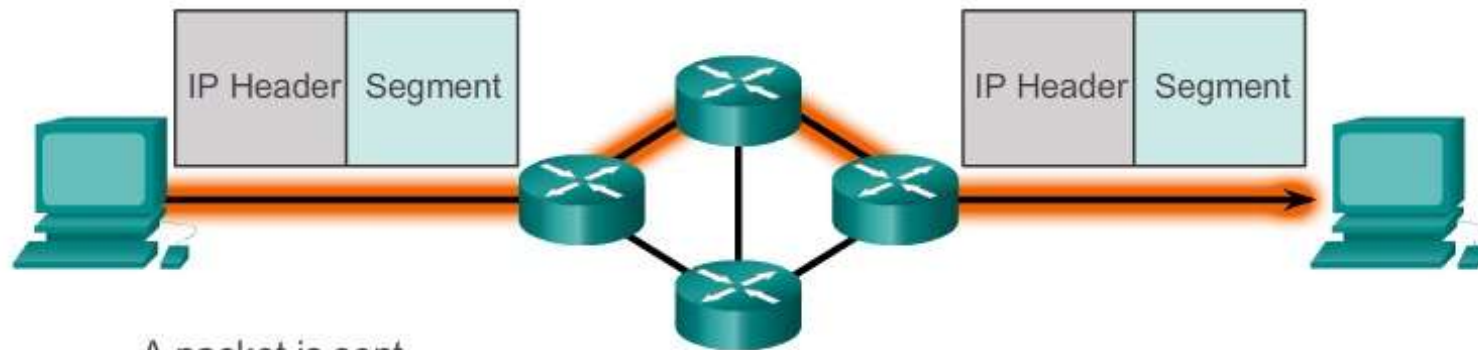
Network Layer

- Two common network layer protocols
 - Internet Protocol version 4 (**IPv4**)
 - Internet Protocol version 6 (**IPv6**)
- The basic characteristics of IP are:
 - **Connectionless** - No connection with the destination is established before sending data packets. It is similar to sending a letter to someone without notifying the recipient in advance
 - **Best Effort (unreliable)** - Packet delivery is not guaranteed. Senders are unaware whether destination devices are on when sending packets.
 - **Media Independent** - Operation is independent of the medium carrying the data.

Network Layer



Connectionless Communication



A packet is sent.

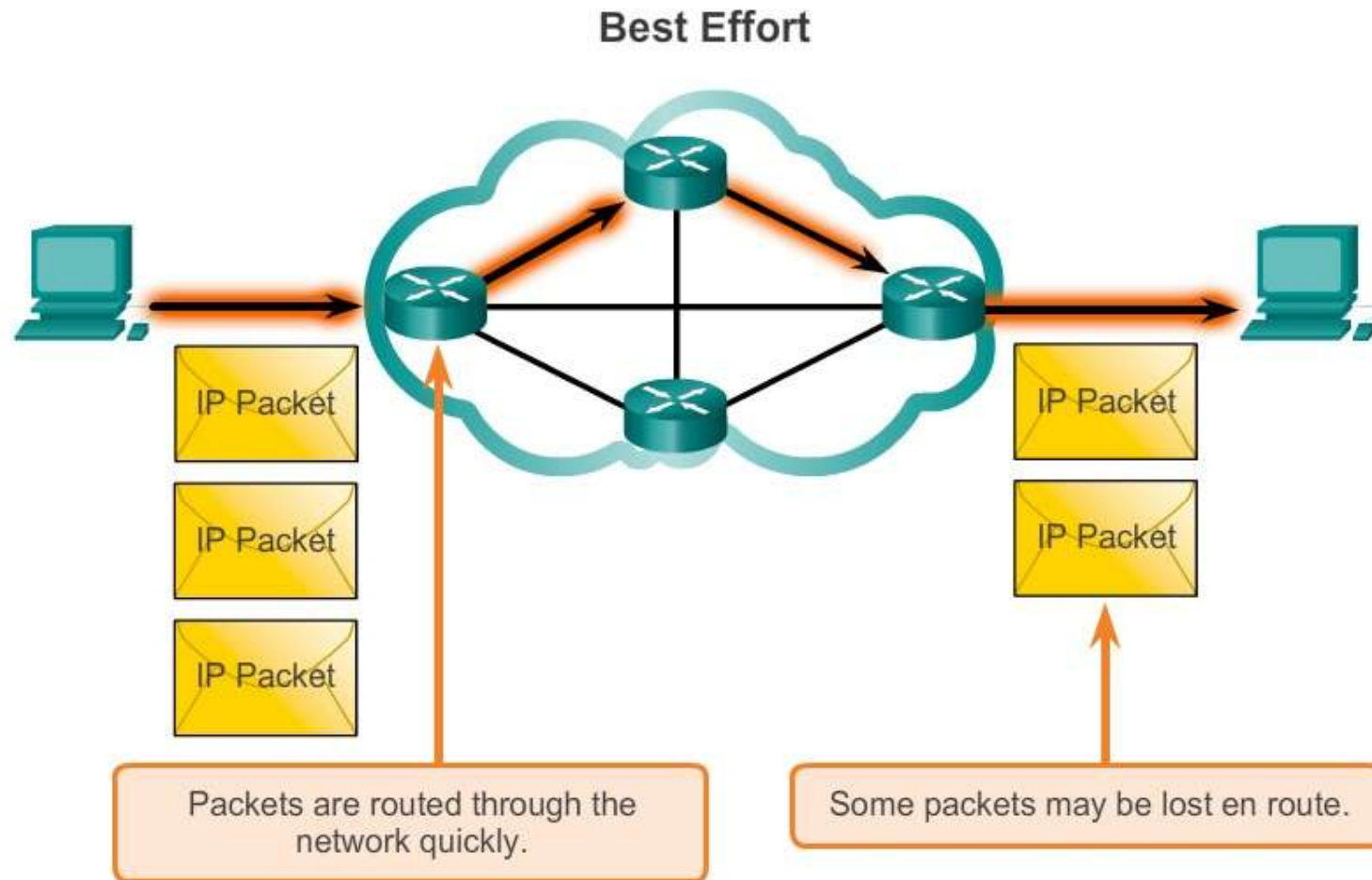
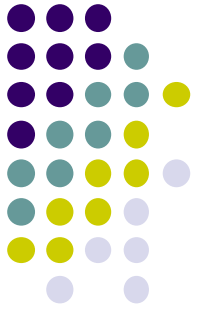
The sender doesn't know:

- If the receiver is present
- If the packet arrived
- If the receiver can read the packet

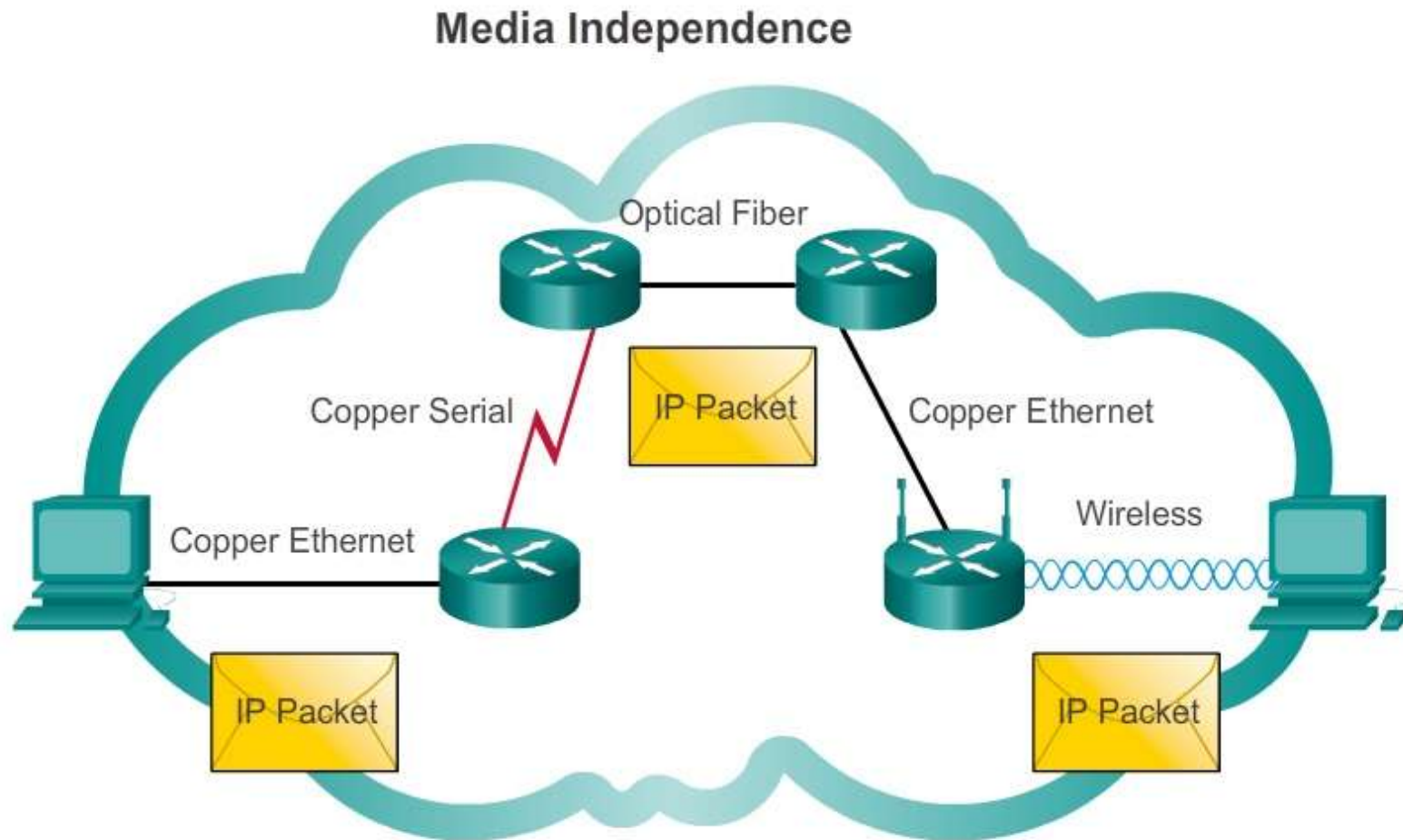
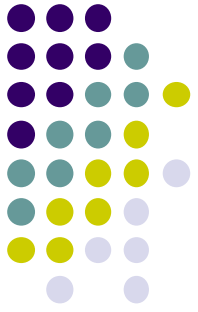
The receiver doesn't know:

- When it is coming

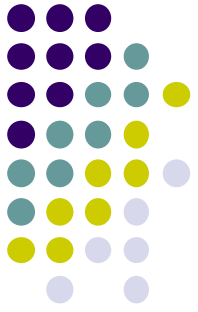
Network Layer



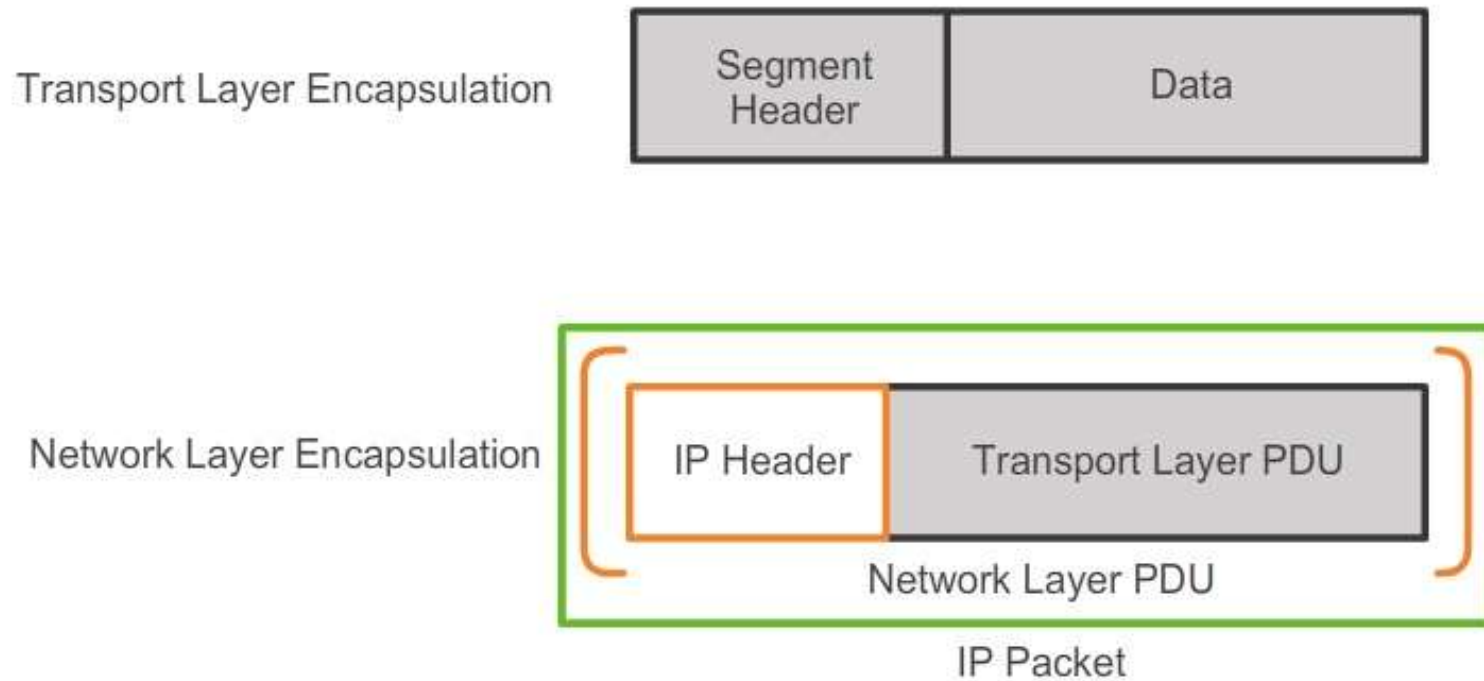
Network Layer



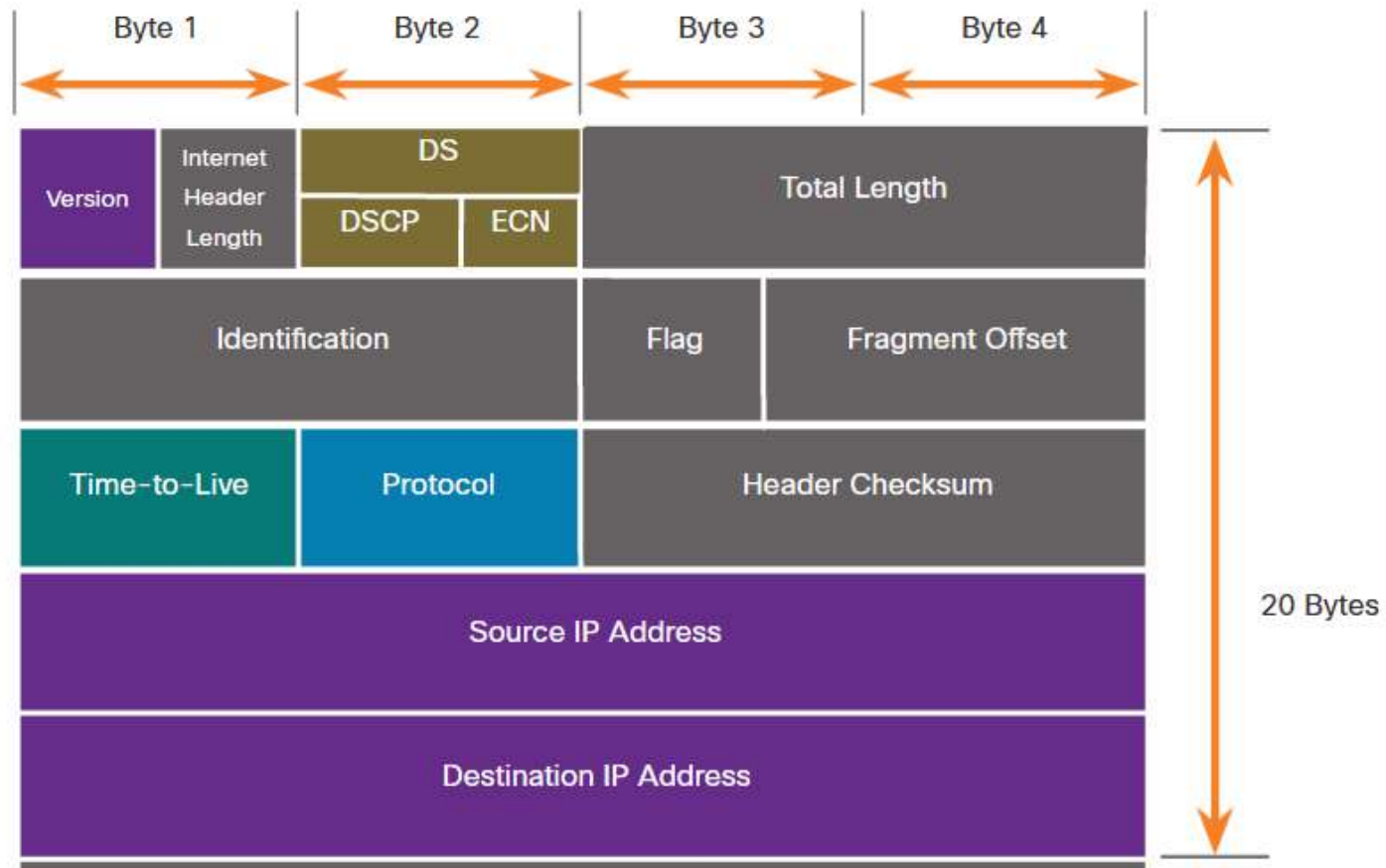
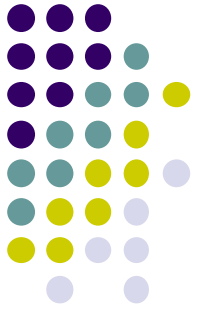
Network Layer



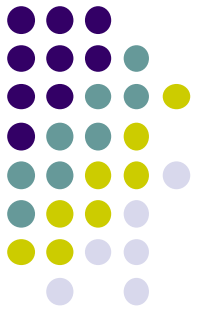
Generating IP Packets



Network Layer

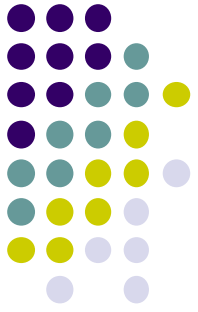


IPv4 Header



Network Layer

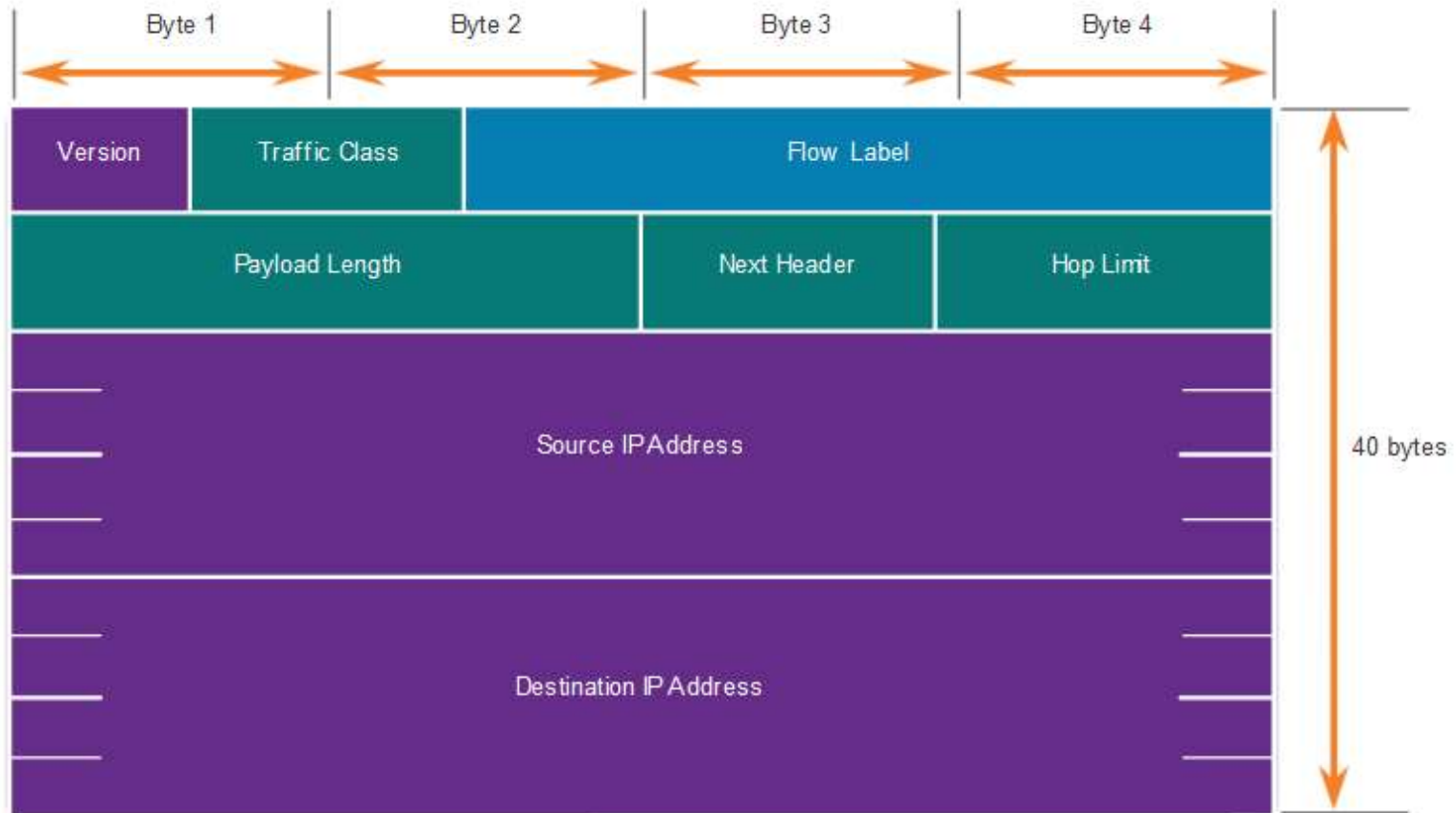
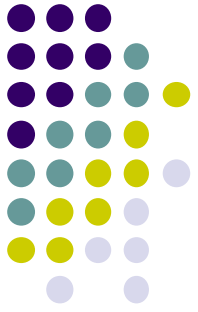
- IPv4 has three major issues:
 - **IP address depletion** - IPv4 has a **limited number** of unique public IP addresses available.
 - **Lack of end-to-end connectivity** - The sharing of public IP address cause an internal **network host is hidden** from outside world. This can be problematic for technologies that require **end-to-end** connectivity.
 - **Increased network complexity** – Network Address Translation in IPv4 creates additional complexity in the network with additional latency and making troubleshooting more difficult.



Network Layer

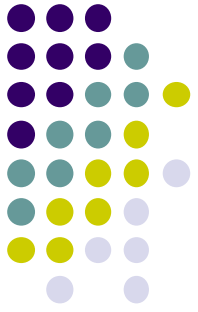
- IPv6 overcomes the limitations of IPv4:
 - **Increased address space** - IPv6 addresses are based on **128-bit**
 - **Improved packet handling** - The IPv6 header has been **simplified** with **fewer fields**.
 - **Eliminates the need for NAT** - With such a large number of public IPv6 addresses, **Network Address Translation (NAT)** is not needed.
 - **Integrated security** - IPv6 natively supports **authentication** and **privacy** capabilities.

Network Layer

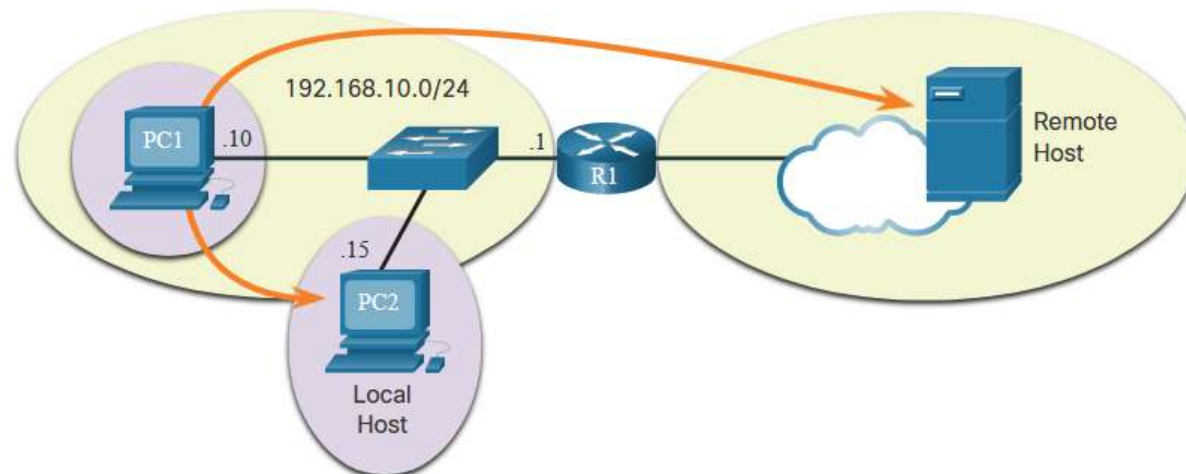


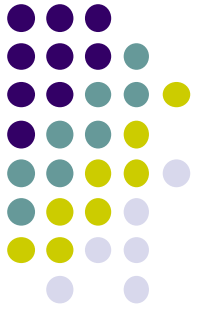
IPv6 Header

Routing



- A host can send a packet to the following:
 - Itself – ping 127.0.0.1 or ::/1 (loopback interface - tests the TCP/IP protocol stack on the host)
 - Local host - This is a **destination host** that is on the same local network as the **sending host**.
 - Remote host - This is a **destination host** on a remote network.

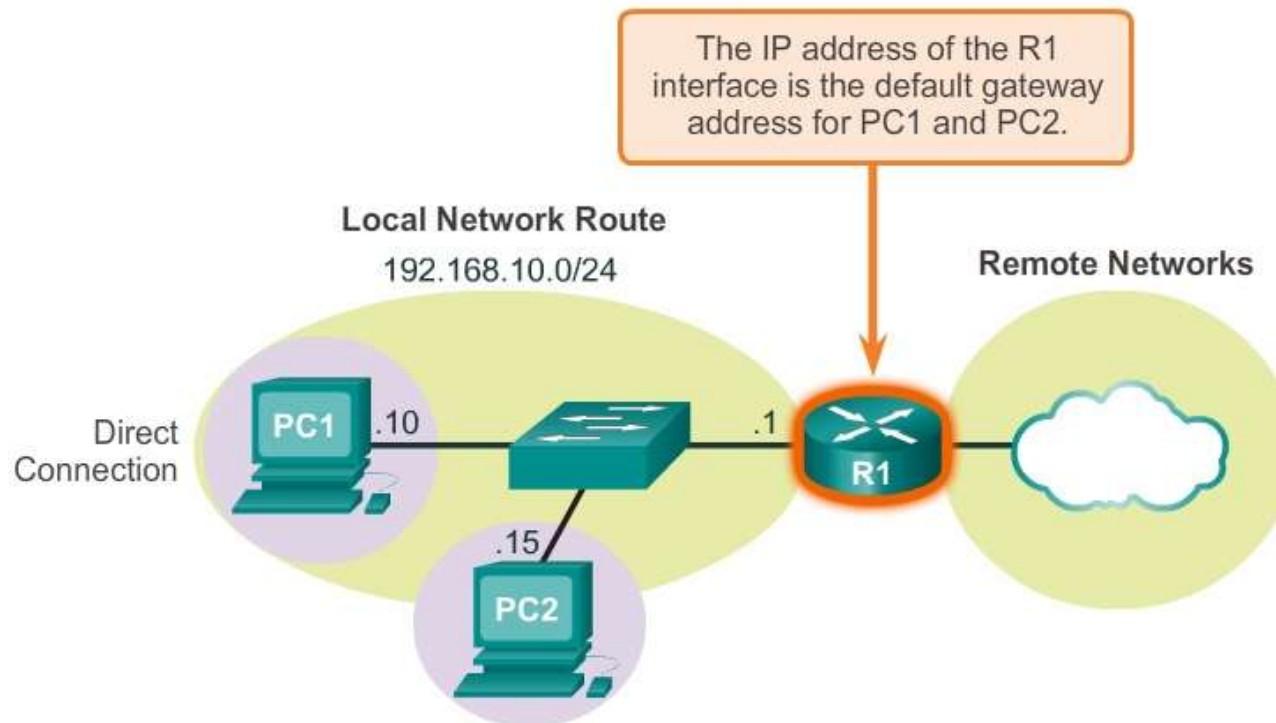
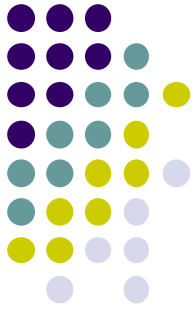




Routing

- The host determines whether the destination IP address is on the same network by
 - Checking its own subnet mask along with its own IPv4 address and the destination IPv4 address. (IPv4)
 - In IPv6 - The local router advertises the local network address (prefix) to all devices on the network.
- If the host is sending a packet to a device on a different IP network, then the host must forward the packet through the intermediate device to the default gateway. The **default gateway** is the device that routes traffic from the local network to devices on remote networks.

Routing



IPv4 Routing Table for PC1

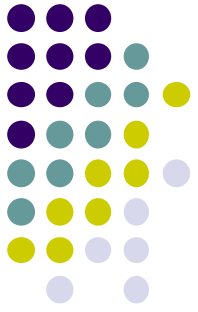
```
C:\Users\PC1> netstat -r
```

```
IPv4 Route Table
```

```
=====
```

```
Active Routes:
```

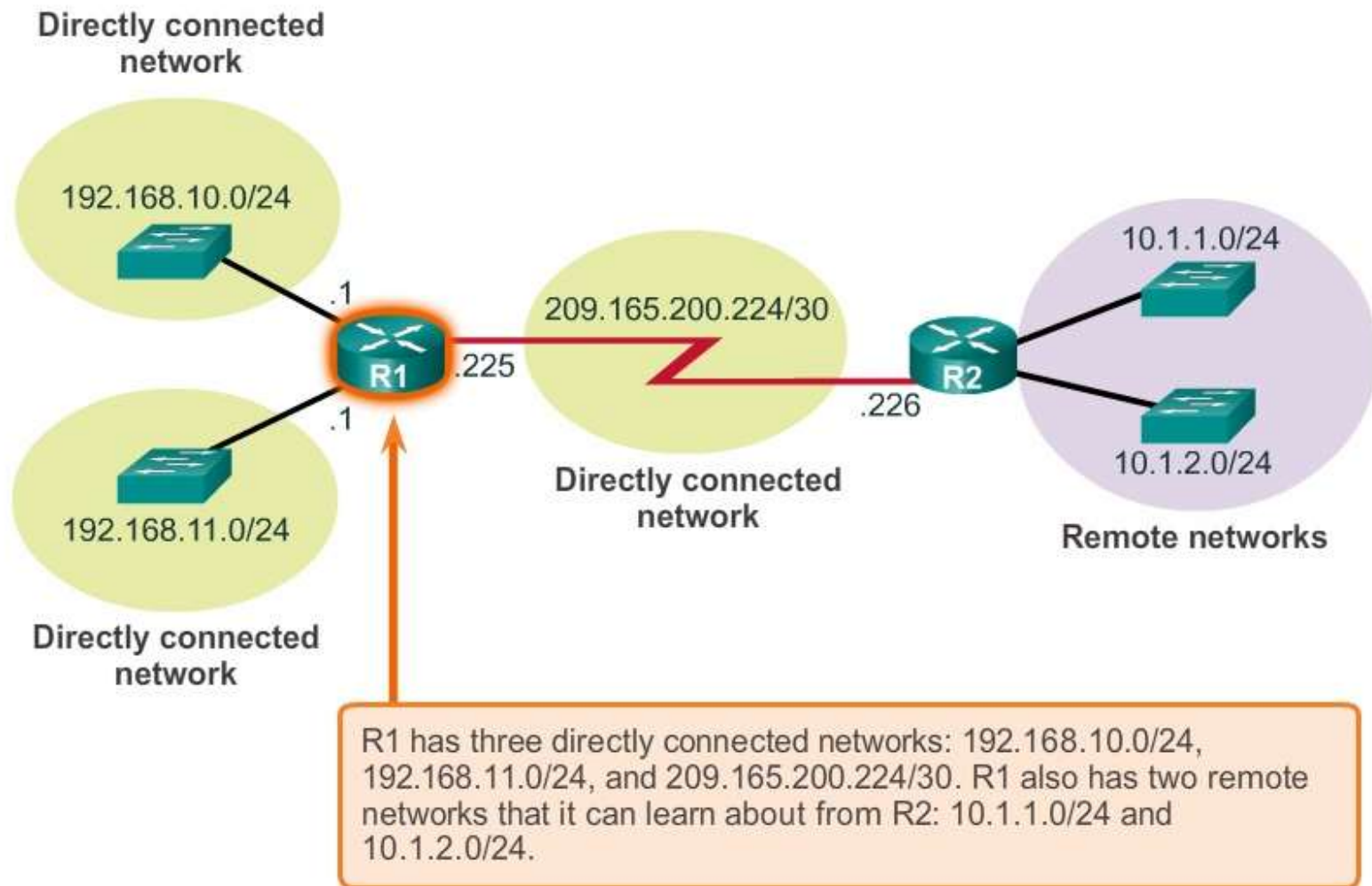
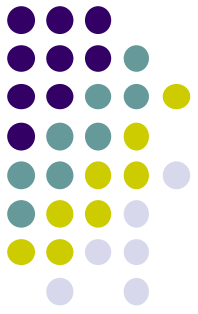
Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.10	25

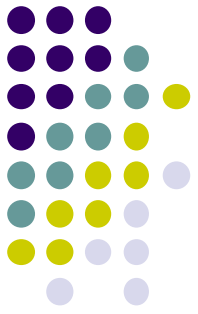


Routing

- Router is the device that connects to multiple networks. Each router contains the routing table that stores three types of route entries:
- **Directly-connected networks** - These network route entries **are active router interfaces**.
- **Remote networks** - These network route entries are connected to other routers. Routers learn about remote networks either by **manual configuration or by exchanging route** information using a **dynamic** routing protocol.
- **Default route** - The default route is used when there is no better (longer) match in the IP routing table.

Routing

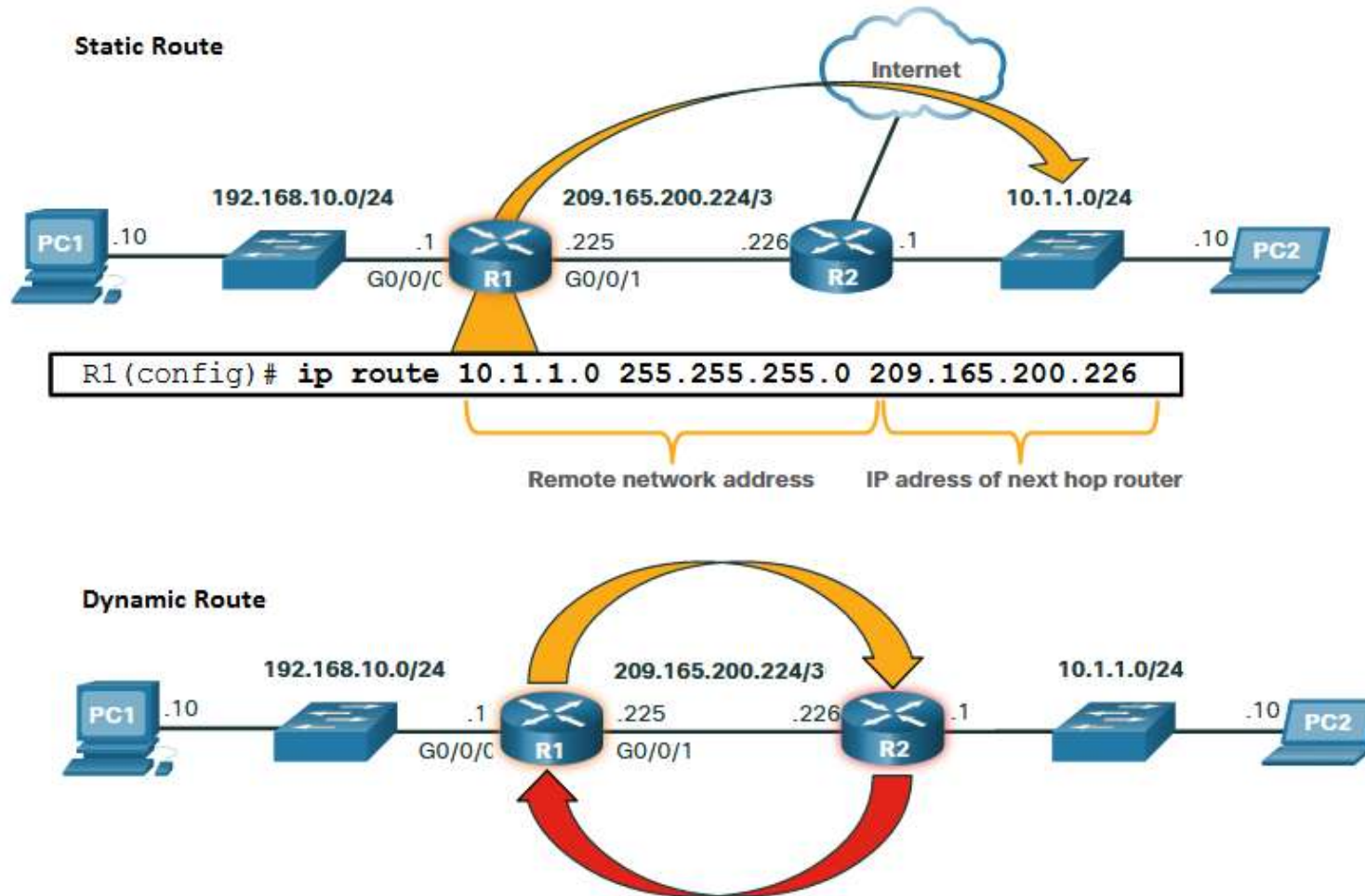
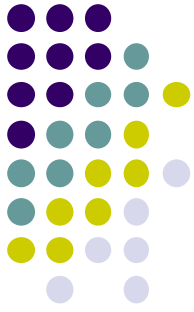




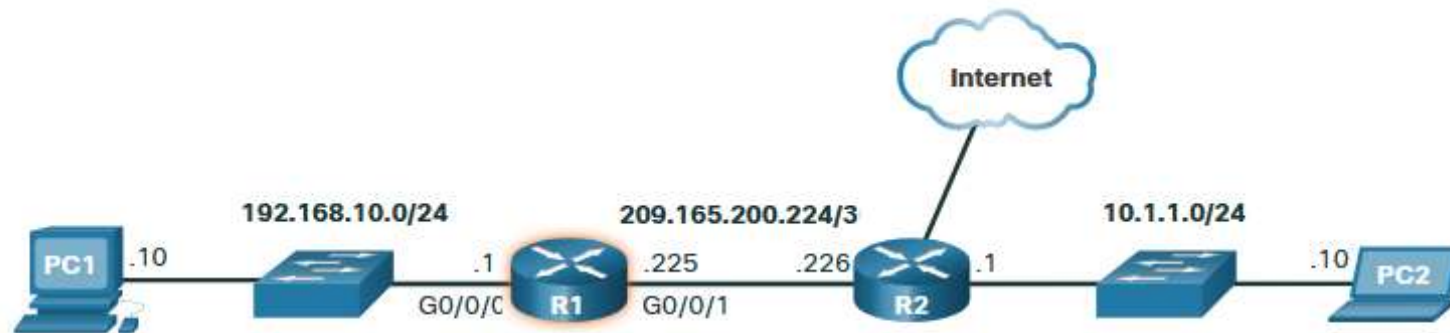
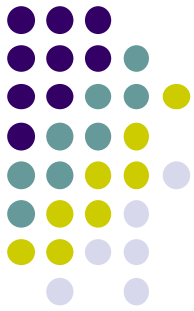
Routing

- Static routes are route entries that are manually configured. If there is a change in the network topology, the static route is not automatically updated and must be **manually reconfigured**.
- A dynamic routing protocol allows the routers to automatically learn about remote networks, including a default route, from other routers. If there is a change in the network topology, the routers will **automatically adjust** and attempt to find a new best path.

Routing



- R1 is using the routing protocol OSPF to let R2 know about the 192.168.10.0/24 network.
- R2 is using the routing protocol OSPF to let R1 know about the 10.1.1.0/24 network.



```
R1# show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0

```
S* 0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
```

10.0.0.0/24 is subnetted, 1 subnets

```
O 10.1.1.0 [110/2] via 209.165.200.226, 00:02:45, GigabitEthernet0/0/1
```

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks

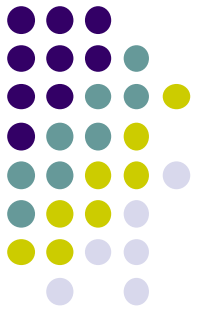
```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
```

```
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
```

209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks

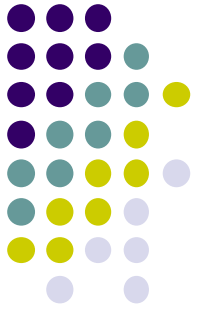
```
C 209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
```

```
L 209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
```



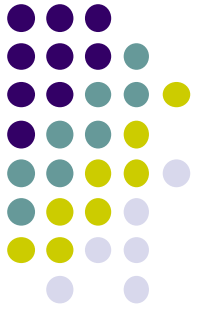
ICMPv4

- An ICMP Echo Message can be used to test the reachability of a host on an IP network. The local host sends an **ICMP Echo Request** to a host. If the host is available, the destination host responds with an **Echo Reply**. (ping command)
- When a host or gateway receives a packet that it cannot deliver, it can use an **ICMP Destination Unreachable** message to notify the source that the destination or service is unreachable.
- An **ICMPv4 Time Exceeded** message is used by a router to indicate that a packet cannot be forwarded because the Time to Live (TTL) field of the packet was decremented to 0. (routing loop)



ICMPv4

- Ping is the command the used for connectivity test. Type of connectivity tests performed include the following:
 - Pinging the local loopback
 - Pinging the default gateway
 - Pinging the remote host
- Traceroute (tracert) is a utility that generates a list of hops that were successfully reached along the path. This list can provide important verification and troubleshooting information. If the data reaches the destination, then the trace lists the interface of every router in the path between the hosts.



ICMPv4

- Traceroute sends three separate packets. The round-trip time (RTT) is the time a packet takes to reach the remote host and for the response from the host to return. An asterisk (*) is used to indicate a lost or unreplied packet.

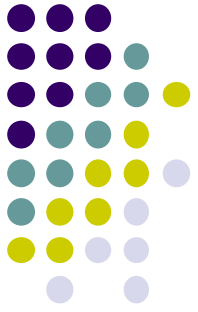
```
C:\Users\Admin>tracert google.com

Tracing route to google.com [216.58.196.206]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    192.168.1.100
  1  12 ms     8 ms     5 ms     103.62.239.241
  2  *          4 ms     7 ms     172.22.22.37
  3  6 ms      7 ms     4 ms     172.22.22.1
  4  12 ms     12 ms    20 ms    45.120.248.10
  5  9 ms      5 ms     8 ms     108.170.251.113
  6  13 ms     12 ms    22 ms    216.239.56.253
  7  3 ms      3 ms     3 ms     del03s06-in-f14.1e100.net [216.58.196.206]

Trace complete.

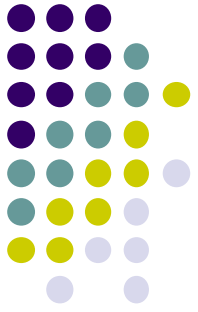
C:\Users\Admin>
```



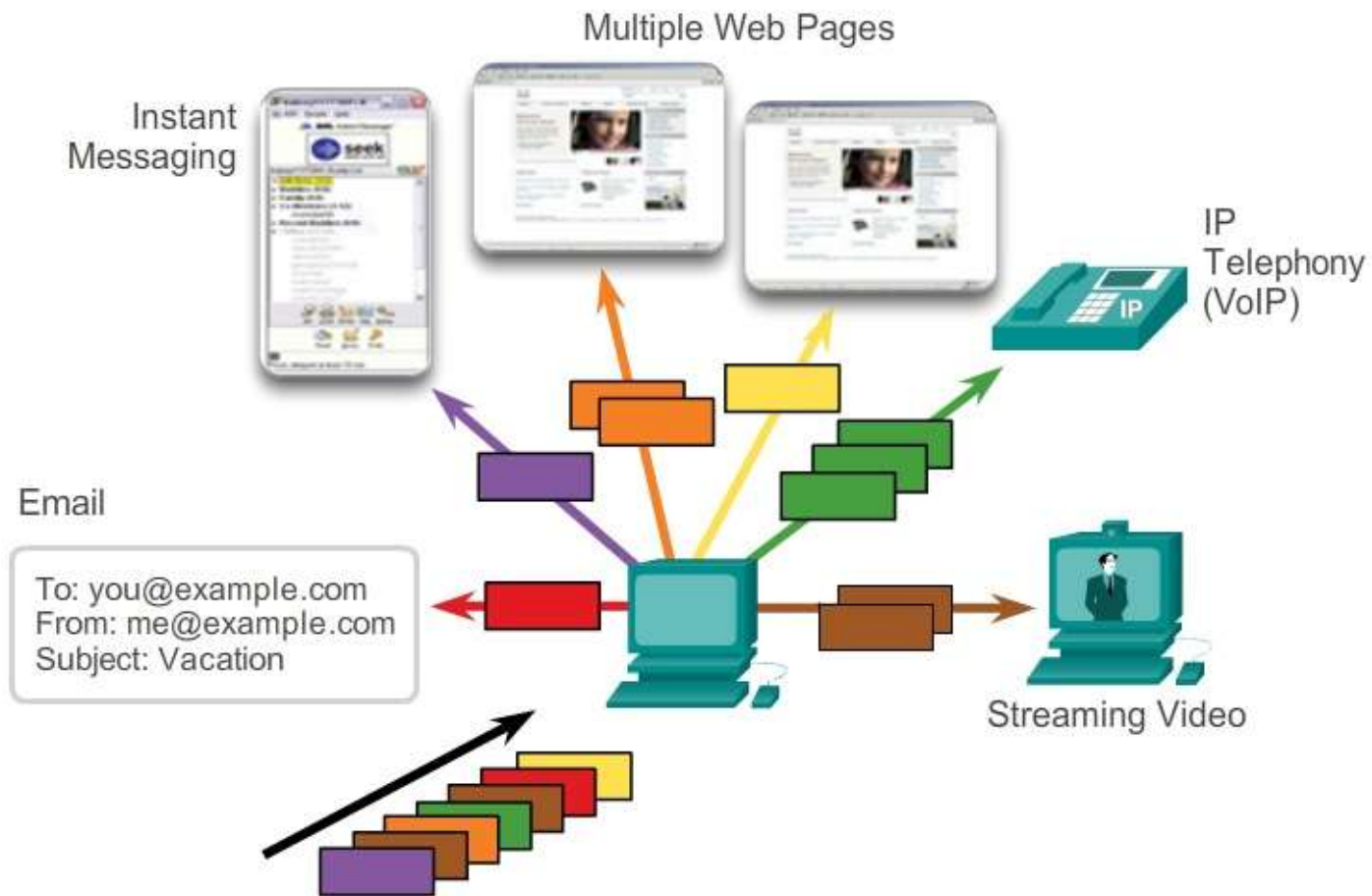
Transport Layer

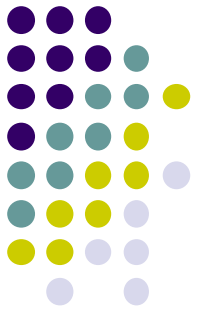
- The transport layer is responsible for establishing a temporary communication session between two applications and delivering data between them.
- The transport layer provides for the segmentation of data, and the controls necessary to reassemble these segments into the various communication streams.
- The transport layer identifies the proper application for each communication stream.
- TCP is considered a reliable transport protocol.
 - Tracking transmitted data segments
 - Acknowledging received data
 - Retransmitting any unacknowledged data

Transport Layer



Identifying the Application

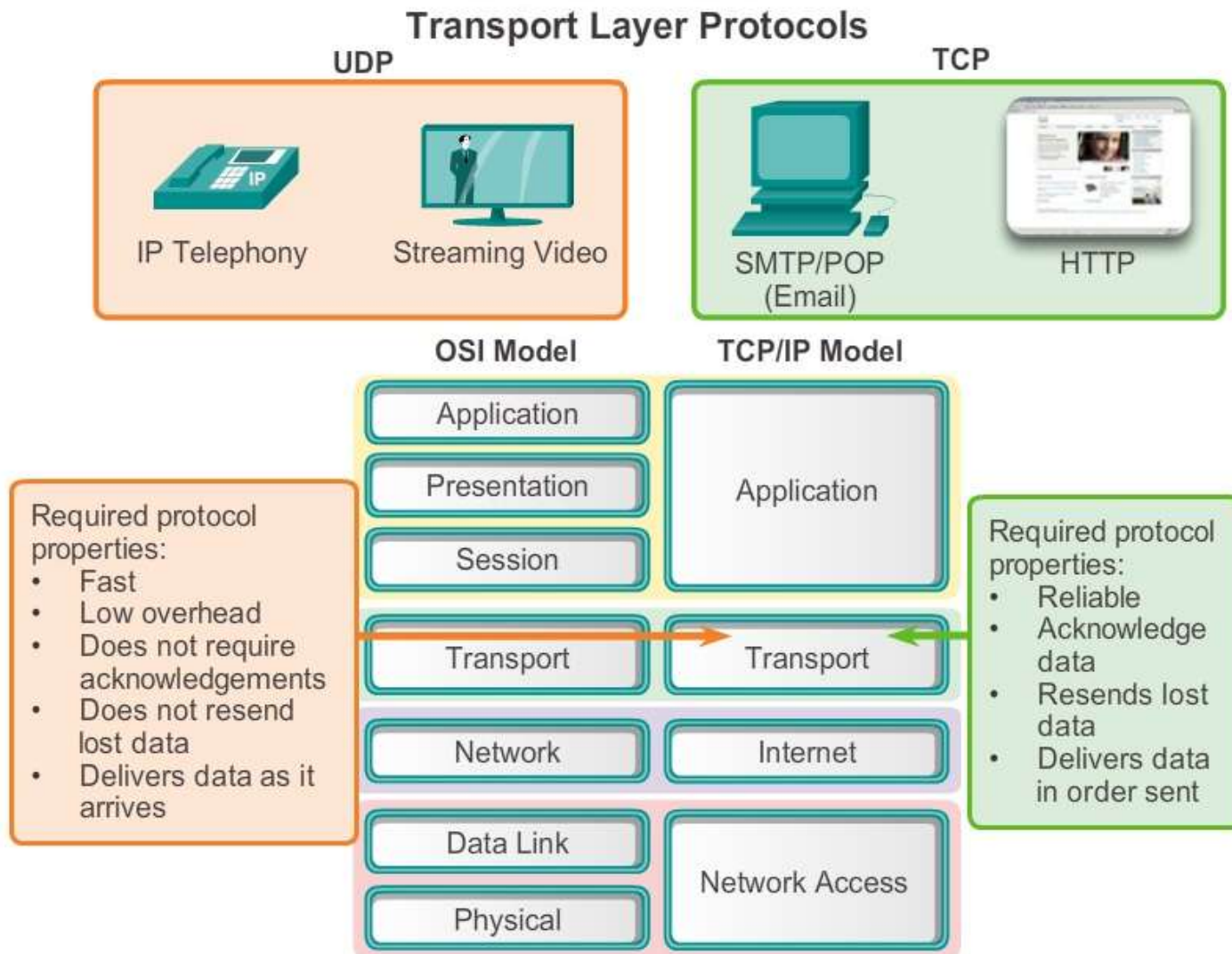
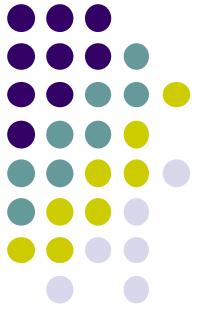


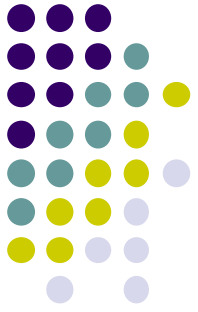


Transport Layer

- UDP provides just the basic functions for delivering data segments between the appropriate applications, with **very little overhead** and data checking.
- Both TCP and UDP are valid transport protocols. Depending upon the application requirements, either one, or sometimes both, of these transport protocols can be used.
- For some applications, segments must arrive in a very specific sequence to be processed successfully. With other applications, all data must be fully received before any of it is considered useful. In other cases, an application can tolerate some data loss during transmission over the network, but delays in transmission are unacceptable.

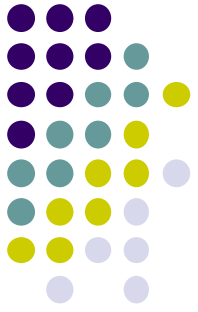
Transport Layer





Transport Layer

- When a message is delivered using either TCP or UDP, the protocols and services requested are identified by a port number.
- A **port number** is a numeric identifier within each segment that is used to keep track of specific conversations and destination services requested.
- **Destination Port**
 - The client places a destination port number in the segment to tell the destination server what service is being requested. For example, port 80 refers to HTTP or web service.



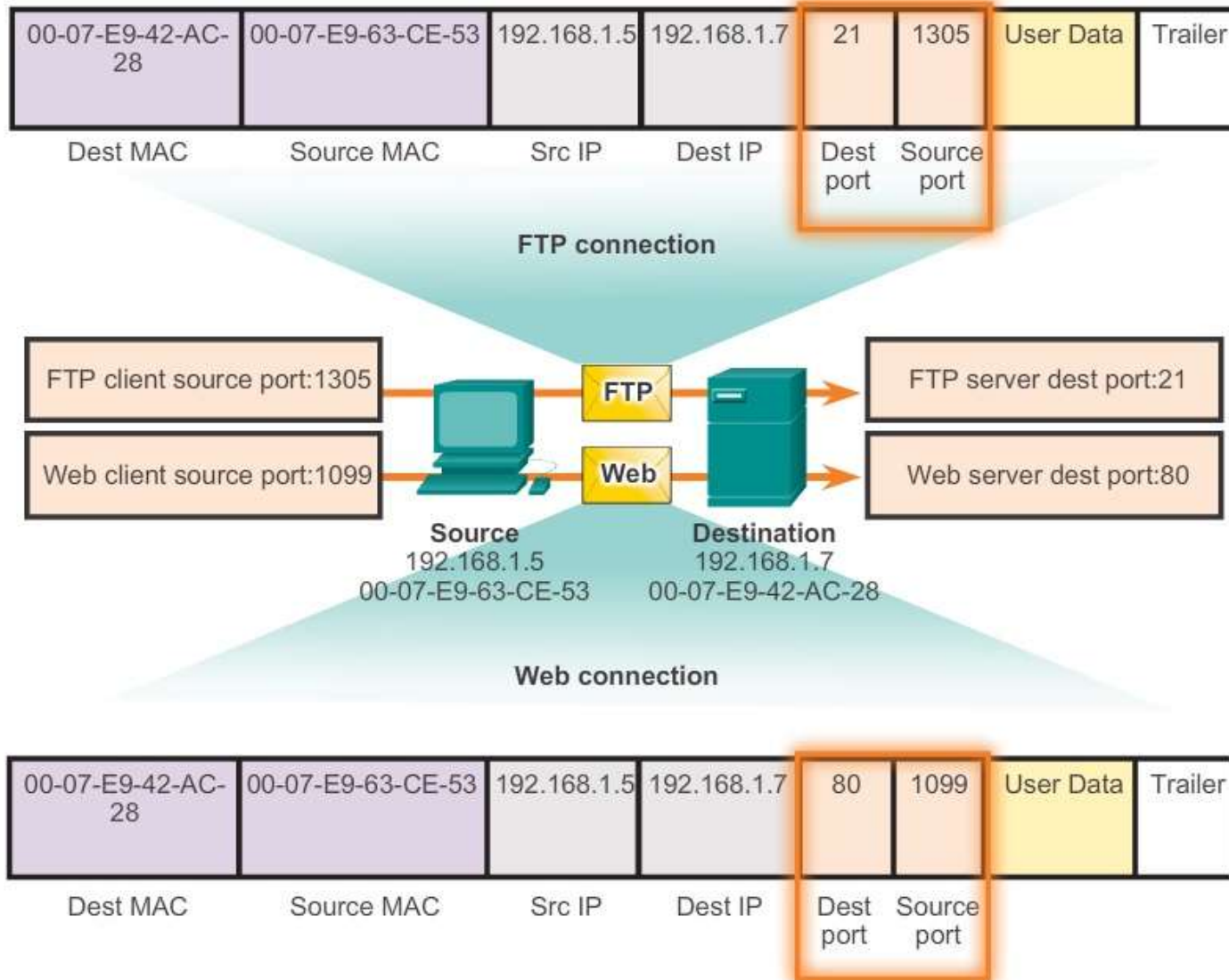
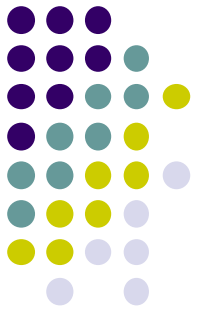
Transport Layer

- **Source Port**
 - The source port number is **randomly generated** by the sending device to identify a conversation between two devices. This allows multiple conversations to occur simultaneously.
- The source and destination ports are placed within the segment. The segments are then encapsulated within an IP packet.

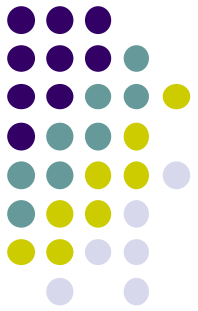
Port Numbers

Port Number Range	Port Group
0 to 1023	Well-known Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

Transport Layer



Transport Layer

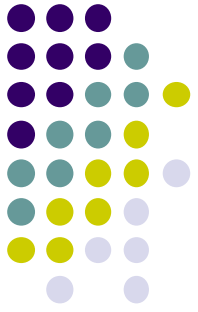


```
C:\> netstat
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	kenpc:3126	192.168.0.2:netbios-ssn	ESTABLISHED
TCP	kenpc:3158	207.138.126.152:http	ESTABLISHED
TCP	kenpc:3159	207.138.126.169:http	ESTABLISHED
TCP	kenpc:3160	207.138.126.169:http	ESTABLISHED
TCP	kenpc:3161	sc.msn.com:http	ESTABLISHED
TCP	kenpc:3166	www.cisco.com:http	ESTABLISHED

```
C:\>
```



Transmission Control Protocol (TCP)

- **Establishing a Session**

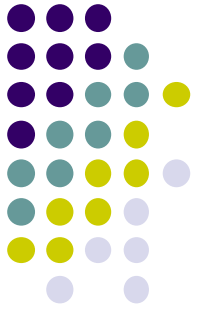
- TCP is a **connection-oriented protocol** that negotiates and establishes a **permanent** connection (or session) between source and destination devices prior to forwarding any traffic.

- **Reliable Delivery**

- TCP can ensure that all pieces reach their destination by having the **source device retransmit lost** or corrupted data.

- **Same-Order Delivery**

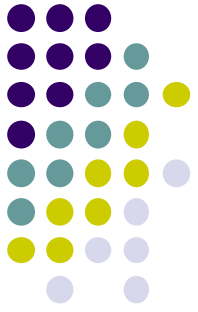
- By numbering and sequencing the segments, TCP can ensure that these **segments** are **reassembled** into the proper order.



Transmission Control Protocol (TCP)

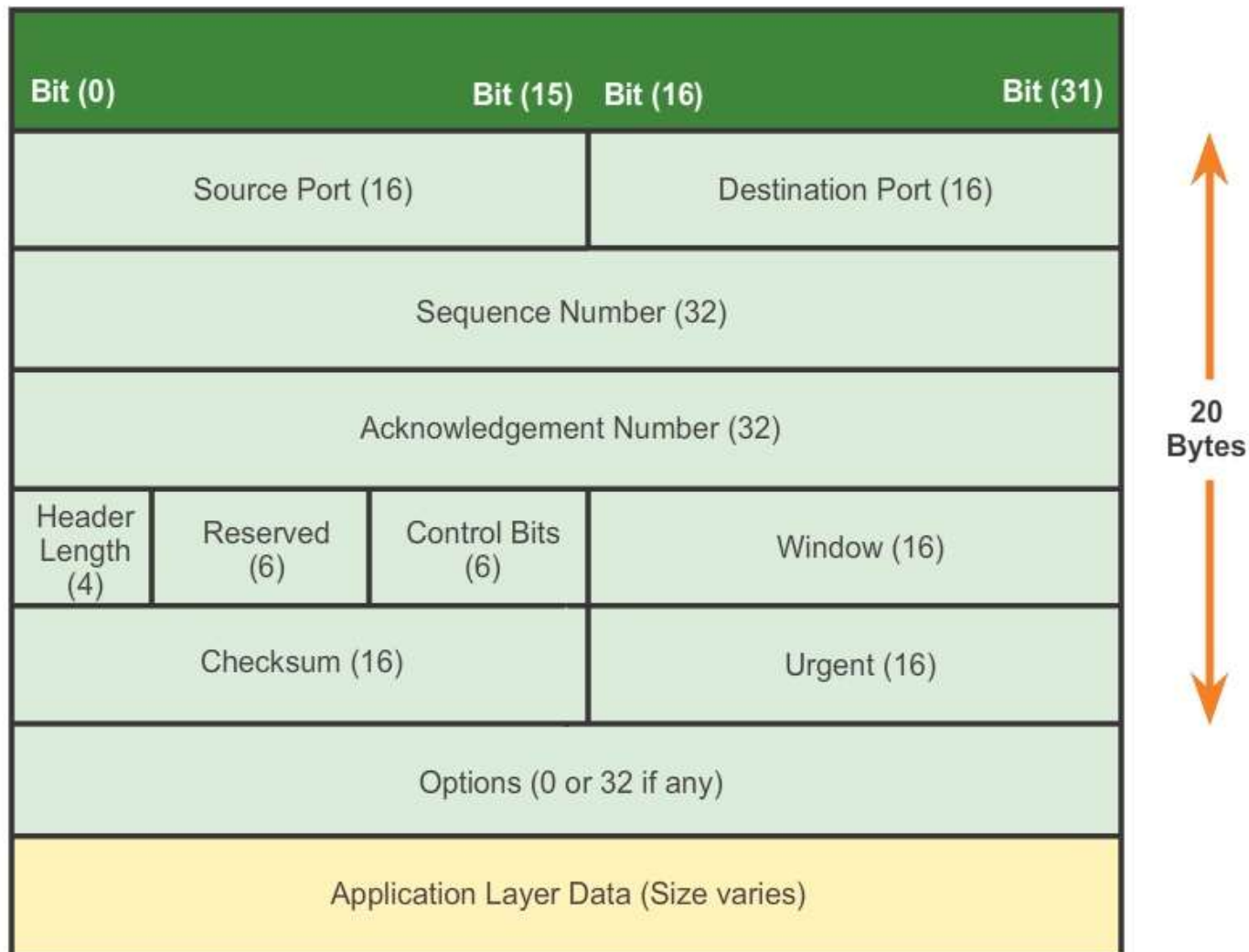
- **Flow Control**

- Network hosts have limited resources, such as memory or bandwidth. When TCP is aware that these resources are overtaxed, it can request that the sending application reduce the rate of data flow.
- Flow control can prevent the loss of segments on the network and avoid the need for retransmission.
- TCP incurs additional overhead. Each TCP segment has **20 bytes of overhead** in the header encapsulating the application layer data. This is considerably more than a UDP segment, which only has 8 bytes of overhead

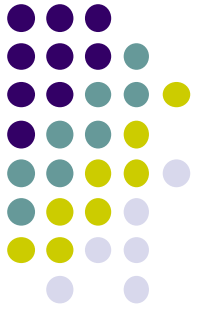


Transmission Control Protocol (TCP)

TCP Segment

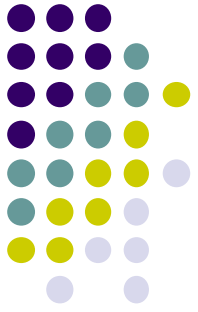


Transmission Control Protocol (TCP)

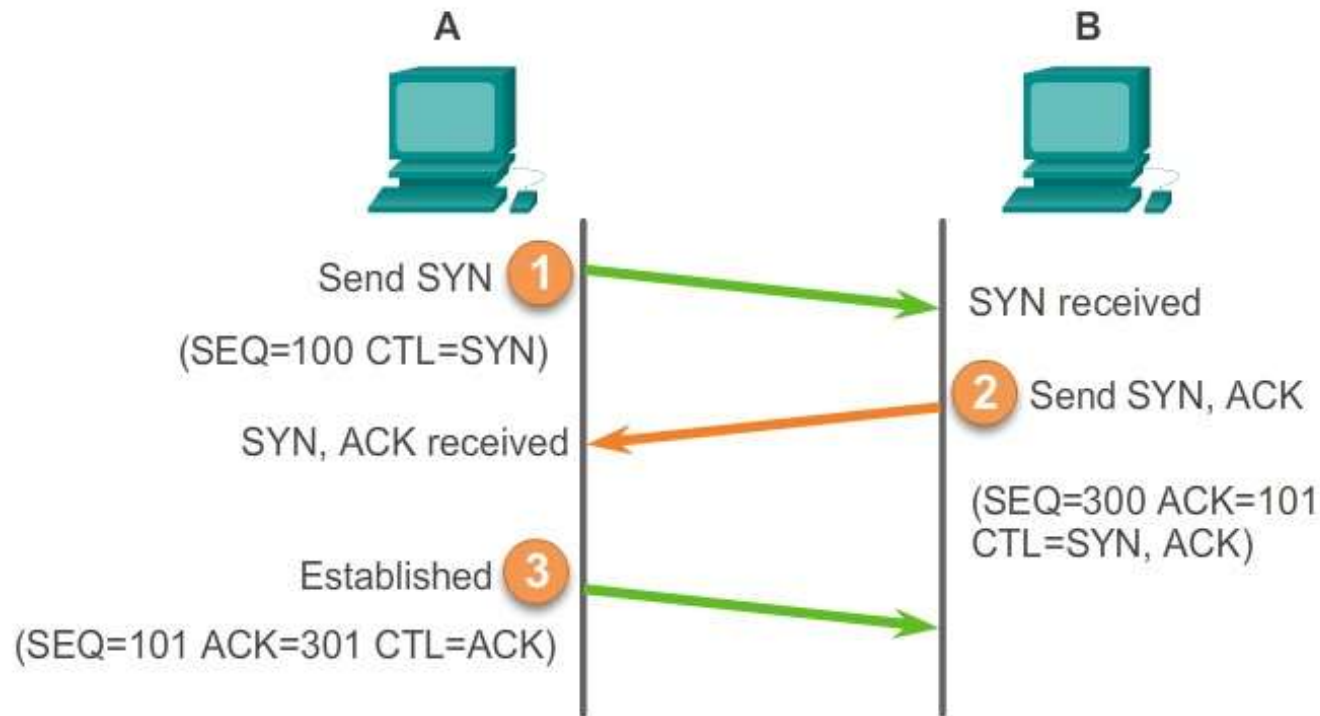


- To establish the connection, TCP perform a three-way **handshake**.
 - A TCP client begins the three-way handshake by sending a segment with the synchronize sequence number (SYN) control flag.
 - The TCP server acknowledge the receipt of the SYN segment from the client to establish the session from the client to the server. The server sends a segment back to the client with the acknowledgement (ACK) flag. The server also send a SYN flag to the client.
 - Finally, the TCP client responds with a segment containing an ACK that is the response to the TCP SYN sent by the server.

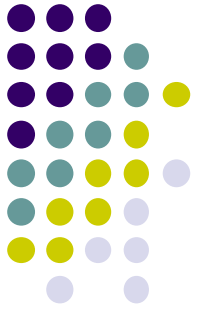
Transmission Control Protocol (TCP)



TCP Connection Establishment

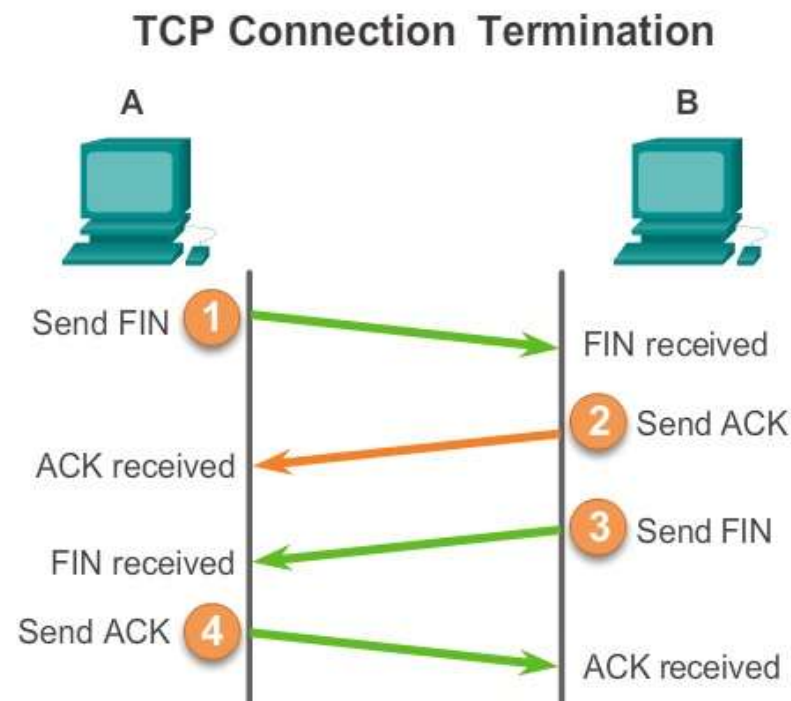


CTL = Which control bits in the TCP header are set to 1
A sends ACK response to B.

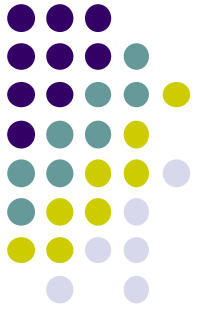


Transmission Control Protocol (TCP)

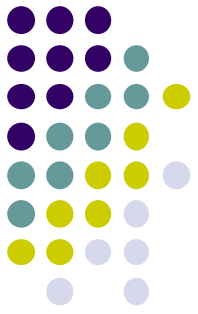
- To close a connection, the **Finish (FIN)** control flag must be set in the segment header. To end each one-way TCP session, a two-way handshake is used, consisting of a FIN segment and an ACK segment.



Transmission Control Protocol (TCP)

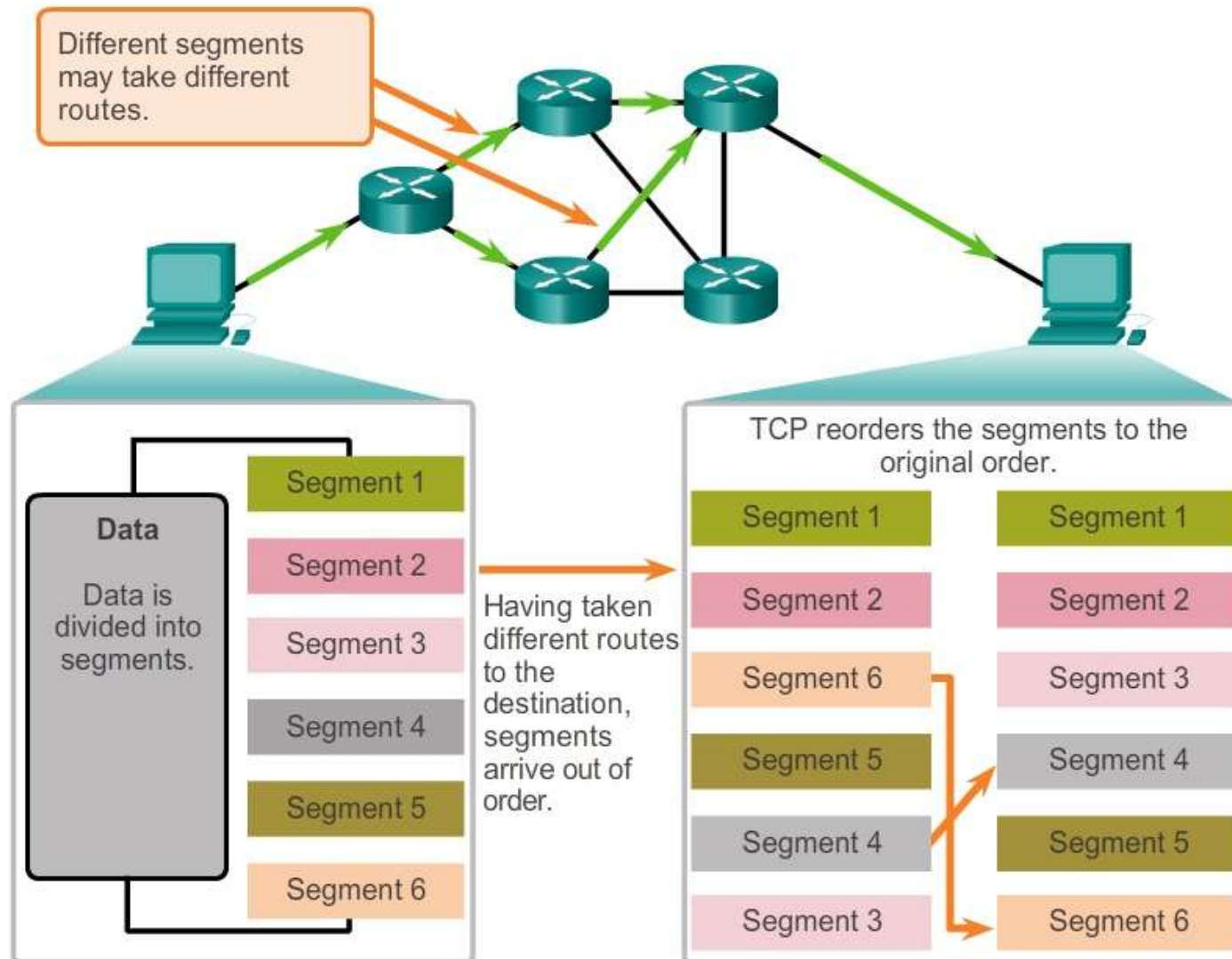


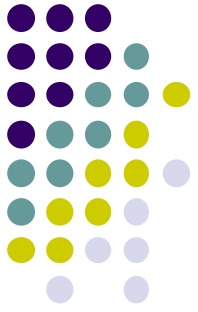
- When services send data using TCP, segments may arrive at their destination out of order.
- The receiving TCP process places the data from a segment into a receiving buffer. Segments are placed in the proper sequence number order and passed to the application layer when reassembled.



Transmission Control Protocol (TCP)

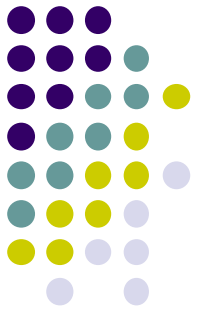
TCP Segments Are Reordered at the Destination





Transmission Control Protocol (TCP)

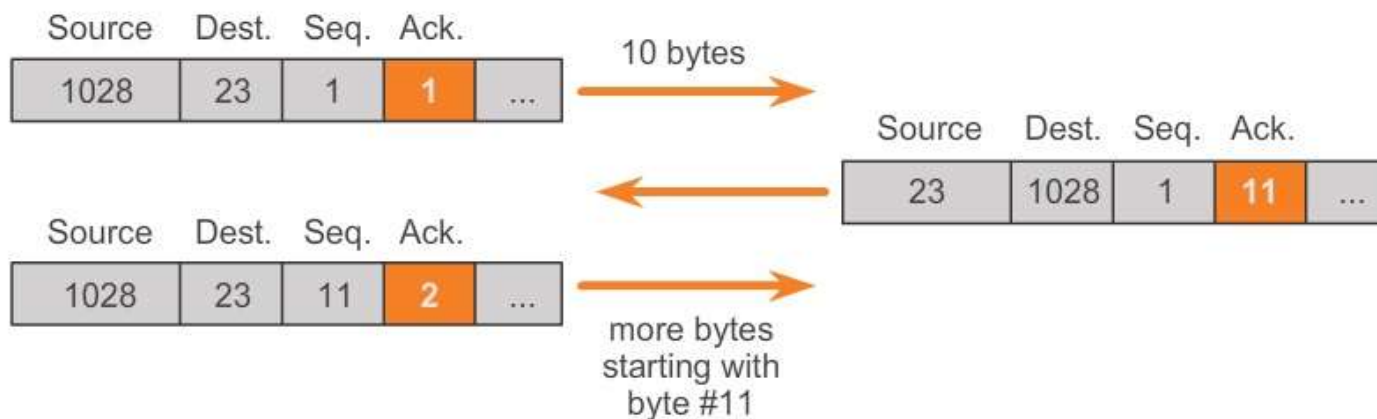
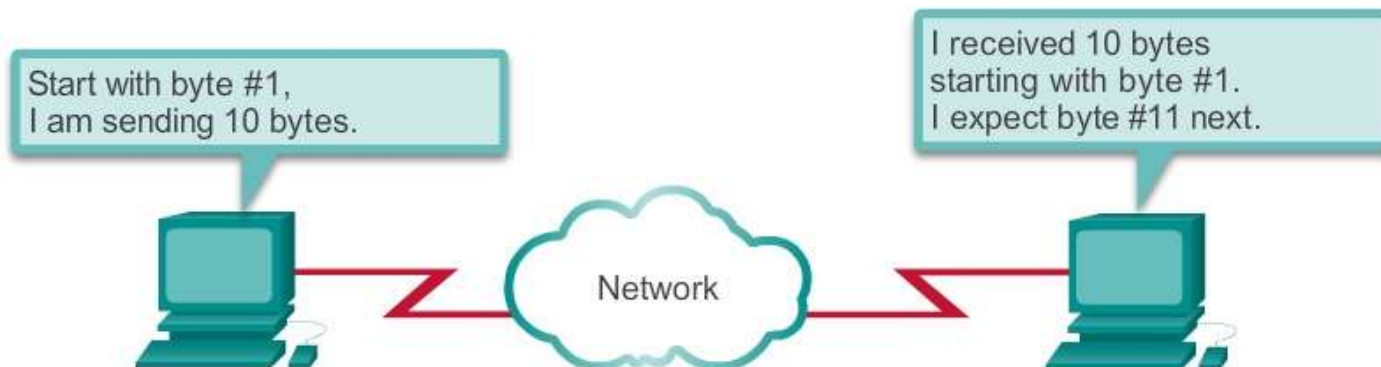
- One of the functions of TCP is ensuring that each segment reaches its destination. The TCP services on the destination host acknowledge the data that it has received by the source application.
- When TCP at the source host has not received an acknowledgement after a predetermined amount of time, it returns to the last ACK number received and retransmits the data from that point forward.
- The amount of data that a source can transmit before an acknowledgement must be received is called the **window size**, which is a field in the TCP header that enables the management of lost data and flow control.



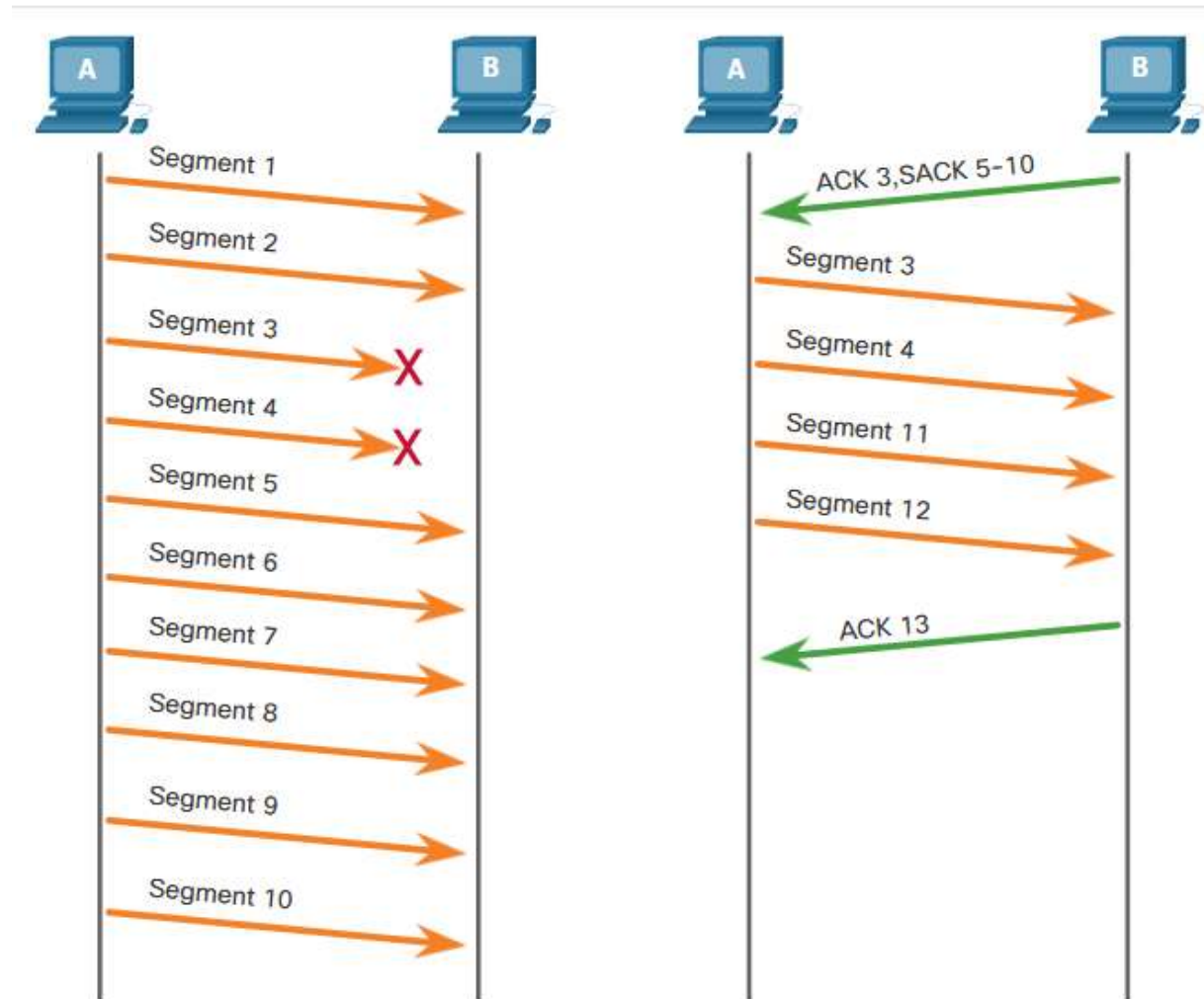
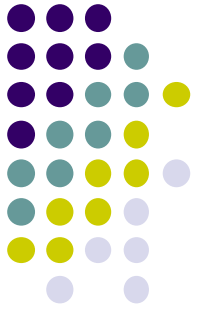
Transmission Control Protocol (TCP)

Acknowledgement of TCP Segments

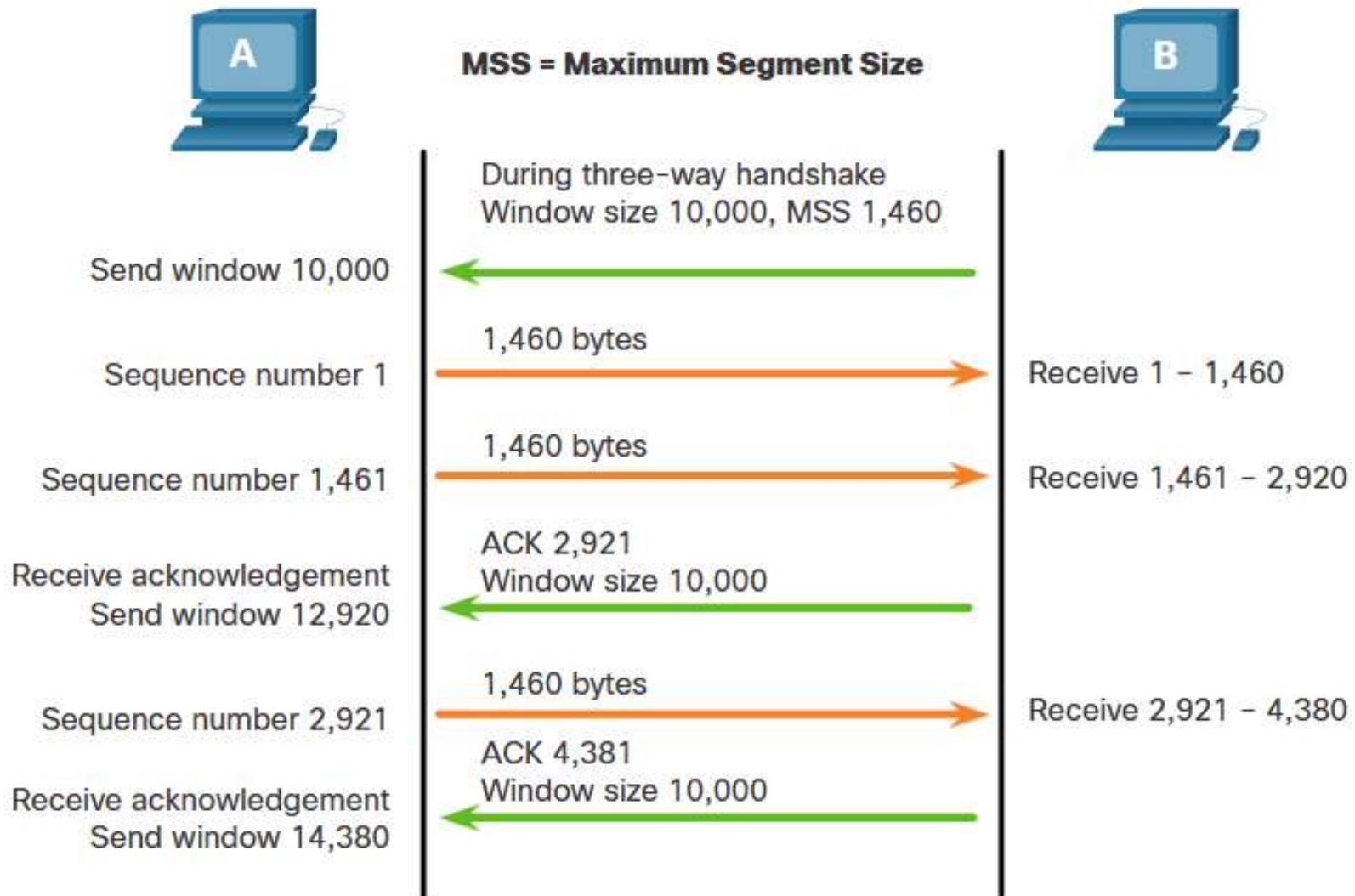
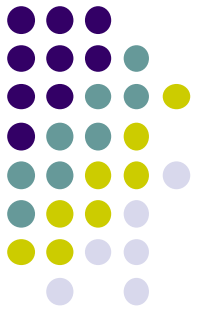
Source Port	Destination Port	Sequence Number	Acknowledgement Numbers	...
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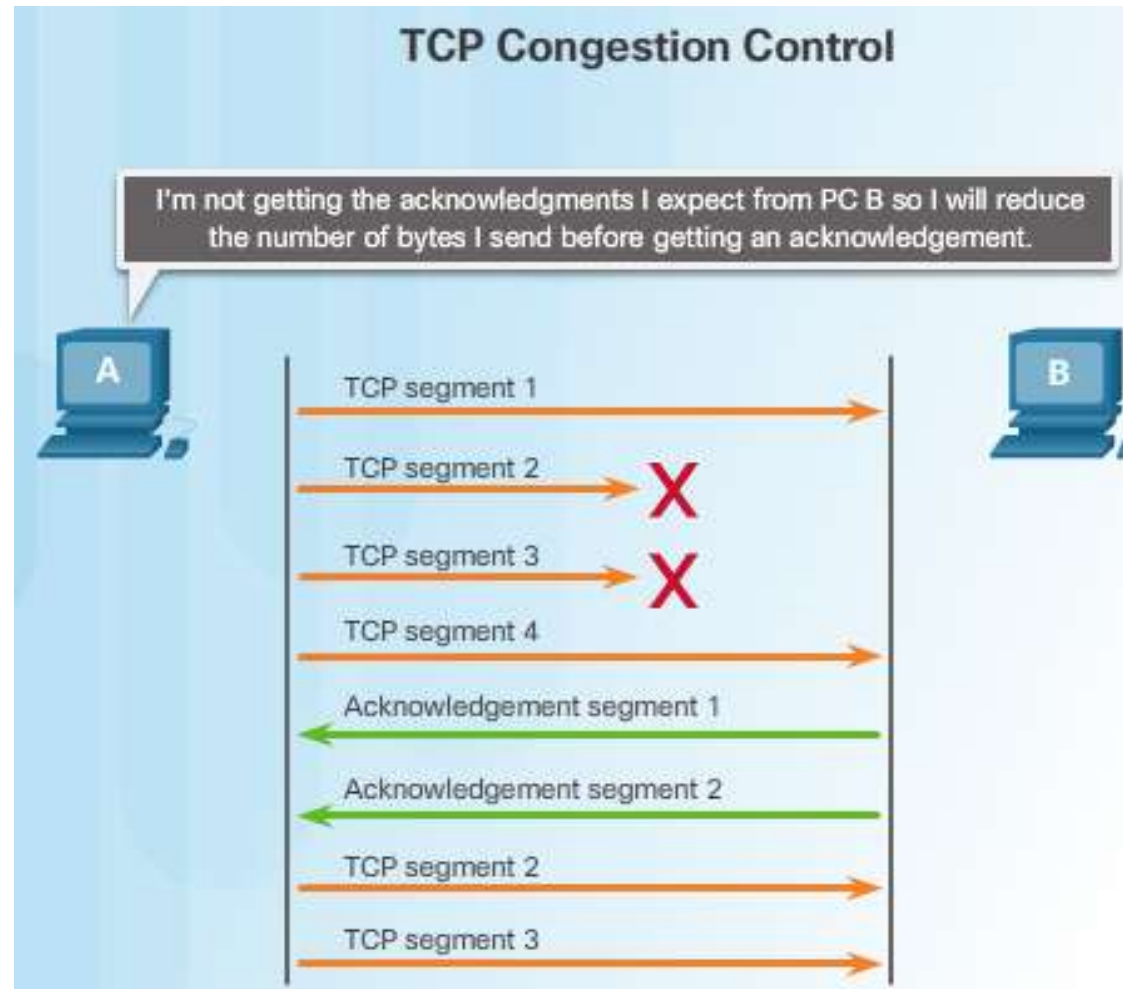
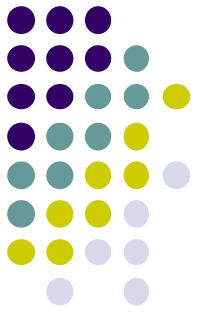
Transmission Control Protocol (TCP)



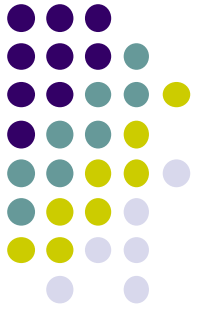
Transmission Control Protocol (TCP)



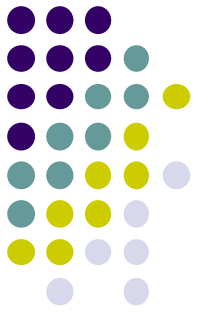
Transmission Control Protocol (TCP)



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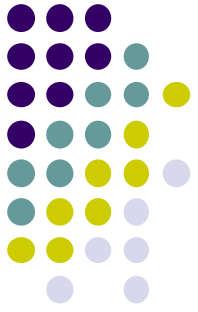


- Some examples of well-known applications that use TCP include:
 - Hypertext Transfer Protocol (HTTP)
 - File Transfer Protocol (FTP)
 - Simple Mail Transfer Protocol (SMTP)
 - Telnet



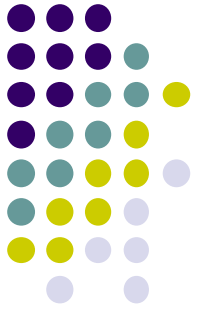
User Datagram Protocol (UDP)

- **Connectionless**
 - UDP does not establish a connection between the hosts before data can be sent and received.
- **Unreliable Delivery**
 - UDP does not provide services to ensure that the data will be delivered reliably.
- **No Ordered Data Reconstruction**
 - Occasionally data is received in a different order than it was sent. UDP does not provide any mechanism for reassembling the data in its original sequence.
- **No Flow Control**
 - There are no mechanisms within UDP to control the amount of data transmitted by the source.



User Datagram Protocol (UDP)

- UDP's low overhead data delivery makes it an ideal transport protocol for applications that can tolerate some data loss.
- The pieces of communication in UDP are called datagrams. These datagrams are sent as best effort by the transport layer protocol.
- A few applications that use UDP are Domain Name System (DNS), video streaming, Voice over IP (VoIP), TFTP and SNMP.



User Datagram Protocol (UDP)

UDP Datagram

